On Single Image Scale-Up Using Sparse-Representations

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Abstract. This paper deals with the single image scale-up problem using sparse-representation modeling. The goal is to recover an original image from its blurred and down-scaled noisy version. Since this problem is highly ill-posed, a prior is needed in order to regularize it. The literature offers various ways to address this problem, ranging from simple linear space-invariant interpolation schemes (e.g., bicubic interpolation), to spatially-adaptive and non-linear filters of various sorts. We embark from a recently-proposed successful algorithm by Yang et. al. [1,2], and similarly assume a local *Sparse-Land* model on image patches, serving as regularization. Several important modifications to the above-mentioned solution are introduced, and are shown to lead to improved results. These modifications include a major simplification of the overall process both in terms of the computational complexity and the algorithm architecture, using a different training approach for the dictionary-pair, and introducing the ability to operate without a training-set by boot-strapping the scale-up task from the given low-resolution image. We demonstrate the results on true images, showing both visual and PSNR improvements.

1 Introduction

Many applications require resolution enhancement of images acquired by low-resolution sensors (e.g. for high-resolution displays), while minimizing visual artifacts. The single image scale-up[1] problem can be formulated as follows: denote the original high-resolution image as $\mathbf{y}_h \in \mathbb{R}^{N_h}$, represented as a vector of length N_h pixels. In addition, denote the blur and decimation operators as $\mathbf{H} : \mathbb{R}^{N_h} \rightarrow \mathbb{R}^{N_h}$ and $\mathbf{S} : \mathbb{R}^{N_h} \rightarrow \mathbb{R}^{N_l}$ (where $N_l < N_h$) respectively. It is assumed hereafter that $\mathbf{H}$ applies a known low-pass filter to the image, and $\mathbf{S}$ performs a decimation by an integer factor s, by discarding rows/columns from the input image. $\mathbf{z}_l \in \mathbb{R}^{N_l}$ is defined to be the low-resolution noisy version of the original image as

$$\mathbf{z}_l = \mathbf{SHy}_h + \mathbf{v}, \tag{1}$$

[1] This problem is often referred to in the literature as *super-resolution*. The term *super-resolution* may be confusing, as it is also used in the context of fusing several low-resolution images into one high-resolution result.[3].

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for an additive i.i.d. white Gaussian noise, denoted by $\mathbf{v} \sim \mathcal{N}\left(0, \sigma^2\mathbf{I}\right)$.

Given $\mathbf{z}_l$, the problem is to find $\hat{\mathbf{y}} \in \mathbb{R}^{N_h}$ such that $\hat{\mathbf{y}} \approx \mathbf{y}_h$. Due to the Gaussianity of $\mathbf{v}$, the maximum-likelihood estimation is obtained by the minimization of $\|\mathbf{SH}\hat{\mathbf{y}} - \mathbf{z}_l\|_2$. However, since $\mathbf{SH}$ is rectangular with more columns than rows, it cannot be inverted stably, and there are infinitely many solutions that lead to a zero value in the above-mentioned least-squares term. Existing single-image scale-up algorithms use various priors on the image in order to stabilize this inversion, ranging from Tikhonov regularization, robust statistics and Total-Variation, sparsity of transform coefficients, and all the way to example-based techniques that use training set of images as priors. While we do not provide a comprehensive review of these techniques, we refer the reader to the following papers [4,5,6,7,8,9,10,11].

In this work we shall use the *Sparse-Land* local model, as introduced in [12,13,14,15], for the scale-up problem. This model assumes that each patch from the images considered can be well represented using a linear combination of few atoms from a dictionary. Put differently, each patch is considered to be generated by multiplying a dictionary by a sparse (mostly zero) vector of coefficients. This assumption will help us in developing an algorithm for image scale-up. It is important to note that this is also the path taken by [1,2] and similar to [16,17]. However, our work differs from their solution in several important aspects, as described in the paper.

This paper is organized as follows: The incorporation of the *Sparse-Land* model into the scale-up problem is shown in Section 2. Section 3 describes the actual implementation details of the algorithm. Experiments and comparative results are given in Section 4, and conclusions are drawn in Section 5.

Just before we embark to the journey of handling the image scale-up problem in this work, we should refer to a delicate matter of the involvement of the blur that operates on the high-resolution image before the sub-sampling. In most cases, when an image is scaled-down, this process is necessarily accompanied by some pre-filter that averages local pixels, in order to reduce aliasing effects. Our work assumes that this is indeed the case, and the role of the scale-up process we aim to develop is to reverse both degradation steps – blur and sub-sampling. As such, our work considers a task that is a generalization of the deblurring problem, and by assuming no sub-sampling, the algorithm we derive here becomes yet-another deblurring method.[2]

There exists a different line of work on the image scale-up problem that strive to separate the treatment of the deblurring and the up-sampling problems. Such work would assume that there is no blur involved, so that the inversion process is purely an interpolation task. Alternatively, if there is a blur, the recovery performance would be measured with respect to the blurred high-resolution image. The rationale of such work is that once the image has been scaled-up in the best possible way, a deblurring stage should be used to get the final outcome. This

[2] In fact, the work reported in [17] does exactly that – developing a deblurring algorithm based on this paradigm.